

SUMMARY TABLE: MATHEMATICS GRADE 10

Sub-Strand 2.1: Similarity and Enlargement — Teacher Reference (Pre-filled)

SUMMARY TABLE: MATHEMATICS GRADE 10	
Sub-Strand	Sub-Strand 2.1: Similarity and Enlargement
Driving Question	DRIVING QUESTION: How can a full-size photograph of a person be resized to fit a small passport photo box without distortion?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Wanjiru's Passport Photo: What Stays the Same When Size Changes?	We observed that Wanjiru's original photo (14 cm × 18 cm), the passport photo (3.5 cm × 4.5 cm) and the poster (70 cm × 90 cm) are very different in size but her face looks the same in each, while the 4.5 cm × 4.5 cm print, made by changing only the width, looks stretched.	We learned that a picture can change size by a large amount and still keep its shape, and that changing only one direction (the width) spoils the shape. We do not yet know exactly what must stay the same.	Our first idea is that the size can change but something about the shape, such as how the sides compare, must stay the same in every direction. We recorded our questions on the Driving Question Board and drew an initial model to test in later lessons.
Lesson 2 Spotting Similar Figures Around Us: Sorting Photos, Maps, Models and Logos	We observed that Wanjiru's 3.5 cm × 4.5 cm passport print and the 70 cm × 90 cm poster look the same shape as the 14 cm × 18 cm original, but the 4.5 cm × 4.5 cm print looks stretched. Other pairs (map, matatu model, school logo) can also be sorted this way.	We learned that similar figures (∼) have the same shape but can have different sizes, and that a figure stretched or squashed in one direction only is not similar to the original.	We can explain that size can change a lot while shape stays the same only when the figure is enlarged or reduced evenly. We still need a reliable test, not just our eyes, which we will build by measuring in Lesson 3.
Lesson 3 Properties of Similar Figures: Equal Angles and Equal Ratios in Wanjiru's Photos	We measured the corner angles, widths, heights, eye gap, chin distance and chin angle of the original (14 cm × 18 cm), passport (3.5 cm × 4.5 cm), poster (70 cm × 90 cm) and the distorted (4.5 cm × 4.5 cm) prints. Record your values: corners __°, width : height __, eye gap : chin distance __, chin angle __°.	Similar figures (∼) have equal corresponding angles and equal ratios of corresponding sides. Write the two tests in your own words and give one number from today's data for each.	Explain why the passport print and poster keep Wanjiru's likeness but the 4.5 cm × 4.5 cm print does not, using the ratio 7 : 9 versus 1 : 1 and the chin angle of about 42.5° versus about 53°.
Lesson 4 Linear Scale Factor: How	We measured and compared the prints of Wanjiru's photo. Original to passport: width	The linear scale factor is $k = \text{image length} \div \text{object length}$, and we must always say	Wanjiru's passport photo ($k = 1/4$) and poster ($k = 5$) keep her likeness because

Many Times Bigger or Smaller? (Wanjiru's Photo Prints)	$3.5 \div 14 = 1/4$ and height $4.5 \div 18 = 1/4$. Original to poster: $70 \div 14 = 5$ and $90 \div 18 = 5$. Passport to poster: 20 for both. Distorted print: width factor $9/28$ and height factor $1/4$, which are not equal.	which is the object and which is the image. If $k > 1$ it is an enlargement and if $0 < k < 1$ it is a reduction. We can find a missing length using $\text{image} = k \times \text{object}$, or $\text{object} = \text{image} \div k$.	every length, both width and height, is multiplied by the same k , so they are similar ($\sim$) to the original. The 4.5×4.5 cm print changed the width by $9/28$ and the height by $1/4$. These are different factors, so her face looked stretched. What must stay the same is one scale factor for all lengths. What may change is the size.
Lesson 5 Similar Triangles in Problem Solving: Measuring Heights with Shadows	What did we observe? For example: Wanjiru (1.5 m tall) casts a 2 m shadow and the tree casts a 12 m shadow at the same time; the angles at the end of both shadows measured about the same; height $\div$ shadow was 0.75 for Wanjiru.	What did we learn? For example: the two shadow triangles are similar (equal corresponding angles, proportional corresponding sides); with Wanjiru's triangle as the object and the tree's as the image, $k = 12 \div 2 = 6$, so the tree is $1.5 \times 6 = 9$ m tall.	How does this help explain the phenomenon? For example: just as the passport photo ($k = 1/4$) and poster ($k = 5$) are similar to the original because every length is multiplied by one scale factor and the angles stay equal, the tree's triangle keeps Wanjiru's triangle's shape; the distorted 4.5×4.5 cm print used different factors ($9/28$ and $1/4$), so its shape changed.
Lesson 6 Area Scale Factor: Why Doubling Lengths Does Not Double Area	We measured and counted unit squares for rectangles enlarged by $k = 2$ and $k = 3$ (areas 24 and 54 cm^2 from a 2×3 rectangle) and calculated the areas of Wanjiru's prints: original $14 \times 18 = 252 \text{ cm}^2$, passport $3.5 \times 4.5 = 15.75 \text{ cm}^2$, poster $70 \times 90 = 6300 \text{ cm}^2$.	When every length is multiplied by the linear scale factor k , the area is multiplied by k^2 . The area factors for Wanjiru's prints are $1/16$ (original to passport, $k = 1/4$), 25 (original to poster, $k = 5$) and 400 (passport to poster, $k = 20$).	Length and width are both multiplied by k , so $\text{area} = (k \times \text{width}) \times (k \times \text{height}) = k^2 \times \text{old area}$. Wanjiru's prints keep her likeness because one scale factor is used in every direction (similar figures), so area changes by k^2 . The distorted 4.5×4.5 cm print used different factors ($9/28$ and $1/4$), so it is not similar and the k^2 rule does not apply.
Lesson 7 Volume Scale Factor: From Small Cartons to Large Cartons (k^3)	Small carton $10 \times 6 \times 4$ cm has volume 240 cm^3 ; large similar carton $20 \times 12 \times 8$ cm has volume 1920 cm^3 ; all edges have $k = 2$ (large $\div$ small); $1920 \div 240 = 8$; a $2 \times 2 \times 2$ block uses 8 unit cubes.	When every length of a solid is multiplied by k , the volume is multiplied by k^3 (and area by k^2), so $k = 2$ gives volume factor 8 and $k = 3$ gives 27.	Volume depends on three lengths (length $\times$ width $\times$ height), each multiplied by the same k , so volume is multiplied by $k \times k \times k = k^3$. This extends the idea from Wanjiru's photo: one scale factor in every direction keeps the shape, while area changes by k^2 and volume by k^3 .
Lesson 8 Enlargement with a Centre and a Positive Scale Factor: Building Wanjiru's Poster from Rays	We drew rays from the centre $O(0,0)$ through $A(1,1)$, $B(3,1)$ and $C(1,2)$ and marked $A'(2,2)$, $B'(6,2)$ and $C'(2,4)$ at twice the distance from O . We measured the sides and angles of triangle ABC and triangle $A'B'C'$.	An enlargement needs a centre and a scale factor k , where $k = \text{image length} \div \text{object length}$. With $k = 2$, every image point lies on the ray from O through its object point at twice the distance. The image is similar to the object: corresponding angles are equal and corresponding sides are in the ratio 1 : 2.	Wanjiru's poster keeps her likeness because every point is moved from one centre by the same scale factor ($k = 5$), so ratios and angles stay the same. The distorted 4.5×4.5 cm print used different factors ($9/28$ for width and $1/4$ for height), so it is not an enlargement and her face looked stretched.
Lesson 9 Fractional and Negative	We observed that when triangle ABC with $A(1,1)$, $B(3,1)$, $C(1,2)$ is enlarged from	We learned that a scale factor between 0 and 1 gives a reduction, a negative scale	Wanjiru's passport print (3.5×4.5 cm) is a reduction of her 14×18 cm original with $k =$

Scale Factors: Reduction and Inverted Images	centre O(0,0) with $k = 1/2$, the image A'(0.5,0.5), B'(1.5,0.5), C'(0.5,1) is smaller and on the same side of O; with $k = -2$, the image A'(-2,-2), B'(-6,-2), C'(-2,-4) is larger, inverted and on the opposite side of O. A projector shows an inverted image of a slide.	factor gives an inverted image on the opposite side of the centre, and lengths are multiplied by the size of k . In every case the corresponding angles stay equal (for example 90°) and the ratios of corresponding sides stay equal, so the image is similar to the object.	$1/4$, since both width and height are multiplied by the same factor and the width : height ratio stays 7:9, so her face looks like her. The 4.5×4.5 cm print used different factors ($9/28$ for width and $1/4$ for height), so the shape changed and her face looked stretched. Size can change by any scale factor, even a fraction or a negative one, but shape is kept only if one scale factor is applied to all lengths.
Lesson 10 Finding the Centre and Scale Factor of an Enlargement	We plotted P(2,2), Q(3,2), R(2,3) and their image P'(4,4), Q'(7,4), R'(4,7), joined P to P', Q to Q' and R to R', extended the lines, and saw that all three meet at one point, (1,1). We measured that each image length is 3 times its object length (for example PQ = 1 and P'Q' = 3).	The centre of an enlargement is the point where the lines joining corresponding object and image points meet. The scale factor is $k = \text{image length} \div \text{object length} = 3$. We checked the result with coordinates: P' minus centre = (3,3) = $3 \times (1,1)$, which is $3 \times (\text{P minus centre})$.	Wanjiru's passport photo ($k = 1/4$) and poster ($k = 5$) are enlargements of the original because one centre and one scale factor apply to every point and length, so her face keeps its shape. The distorted 4.5×4.5 cm print used two different factors ($9/28$ for width and $1/4$ for height), so it is not an enlargement from a single centre and her face looked stretched.
Lesson 11 Applications of Scale: Maps, Models and Costing Painted Signs	We observed that Kisumu and Nairobi, about 265 km apart, are 5.3 cm apart on a 1:5 000 000 map ($26\,500\,000 \text{ cm} \div 5\,000\,000$). We observed that the $70 \text{ cm} \times 90 \text{ cm}$ sign (0.63 m^2) enlarged with $k = 3$ becomes $210 \text{ cm} \times 270 \text{ cm}$ (5.67 m^2), 9 times the area, so paint at KES 400 per m^2 costs KES 252 for the small sign and KES 2268 for the large one. We also saw the $k = 2$ carton hold 8 times as much (240 cm^3 to 1920 cm^3).	We learned that map makers, signwriters and model makers use scale factor k , area factor k^2 and volume factor k^3 to resize objects. Costs that depend on area (such as paint) change by k^2 , not k . The same k must be applied to all lengths in every direction.	This helps explain the photograph: Mr Otieno applied one factor ($k = 1/4$ for the passport print, $k = 5$ for the poster) to both width and height, so Wanjiru's face keeps its shape, just as a map keeps the shape of Kenya. The $4.5 \text{ cm} \times 4.5 \text{ cm}$ print used different factors ($9/28$ for width and $1/4$ for height), so it was distorted.
Lesson 12 Consolidation: Resolving the Passport Photo Puzzle	We observed that the original $14 \times 18 \text{ cm}$ print, the passport photo $3.5 \times 4.5 \text{ cm}$ and the poster $70 \times 90 \text{ cm}$ all have a width : height ratio of 7:9 and the same angles (90° corners, about 42.5° at the chin), while the $4.5 \times 4.5 \text{ cm}$ distorted print has ratio 1:1 and a chin angle of about 53° .	We learned that similar figures have equal corresponding angles and equal ratios of corresponding sides, that the linear scale factor $k = \text{image length} \div \text{object length}$ ($1/4$, 5 and 20 for the passport, poster and passport-to-poster), and that area changes by k^2 ($1/16$, 25 and 400) and volume by k^3 .	Wanjiru's photo keeps her likeness because every length was multiplied by the same scale factor ($k = 1/4$ for the passport photo, $k = 5$ for the poster), so ratios and angles stay equal; the $4.5 \times 4.5 \text{ cm}$ print used different factors ($9/28$ for width and $1/4$ for height), so the ratio and the chin angle changed and her face looked stretched. Size can change by any amount, but shape is kept only when one scale factor is applied to all lengths.

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